Recovery of Missing Sensor Data by Reconstructing Time-varying Graph Signals

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Presentation overview

1. Motivation & Challenges
2. Primary Contributions
3. Proposed Algorithm
4. Experimental Framework

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Motivation and Challenges

Wireless Sensor Networks (WSNs) are among the most promising technologies of the current era. Primary reasons for their widespread application include:

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- Low cost
- Small size
- Low power consumption
- Easy to deploy

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Replacing the malfunctioning/ inactive sensors becomes a **tedious task**.

Accurately reconstructing the missing values is the only way forward.

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The proposed method improves upon them to a great extent.

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Primary Contributions

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Graph Signal Processing (GSP) is one such graph-based technique. Sampling and reconstruction of graph signals in GSP is useful for various real-life applications.

In our work, the graph vertices represent the relative position of the sensor nodes and the vertex-indexed signals represent the sensor attributes.

To summarize our contributions:

- Introducing the concept of **reconstructing time-varying graph signals** to **recover missing data** in wireless sensor networks.

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- This work is inspired by the 2020 MLSP paper by Giraldo *et al.*

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- There they **predict the number of new COVID-19 cases** by extending the **Sobolev norm** defined in GSP for **time-varying graph signals**.
- Here we utilize the Sobolev method for the recovery purpose
- The Sobolev algorithms **shows significant performance improvement over SoTA**.
- We validate our claim by performing experiments on **several publicly available datasets** collected from diverse environments (indoor and outdoor).

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Proposed Algorithm

Graph Construction

$G = (\nu, \mathcal{E}, \mathbf{W})$ is a graph with ν as the set of vertices, $\mathcal{E}$ as the set of edges and $\mathbf{W}$ as the weighted adjacency matrix.

$\mathbf{D}$ is the degree matrix and $\mathbf{L} = \mathbf{D} - \mathbf{W}$ is the unnormalized Laplacian matrix of G.

$x : \nu \to \mathbb{R}$ is a vertex-indexed graph signal.

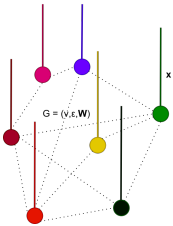

Figure: Graph and graph signals.

Sampling and Reconstruction of Graph Signals

$\boldsymbol{\theta}$ is a sampling operator.

$\mathbf{x}(S) = \boldsymbol{\theta}\mathbf{x}$, S is the sequence of sampled indices.

$\boldsymbol{\phi}$ is the interpolation operator.

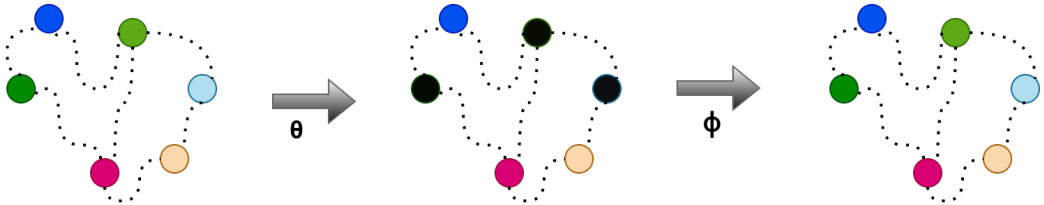

Figure: Graph signals sampling and reconstruction.

Figure: A sensor network.

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Figure: A sensor network.

- We model the missing sensor data recovery problem as the **reconstruction of time-varying graph signals**.
- We perform the reconstruction by **minimizing the Sobolev norm of time-varying graph signals**.

NOTE:

- We have made a prior assumption that the graph signals are smooth in space and time.

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- We have made a prior assumption that the graph signals are smooth in space and time.
- This means that nearby sensors have similar readings and changes in readings are not abrupt with time.

Let $\mathbf{X} = [\mathbf{x_1}, \mathbf{x_2}, \ldots, \mathbf{x_M}]^T$ be a time-varying graph signal, where $\mathbf{x}_t$ denotes the signal at time t $(1 < t < M)$ in G. Here each row of $\mathbf{X}$ represents a time-series on the corresponding vertex.

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According to Qiu *et al.* (2017), we define the smoothness function as:

$$S_2(\mathbf{X}) = \sum_{t=1}^{M} S_2(\mathbf{x_t}) = \mathrm{tr}\left(\mathbf{X}^{\mathrm{T}}\mathbf{L}\mathbf{X}\right).$$

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To include the temporal information, we also define the temporal difference operator as:

$$\mathbf{D}_h = \begin{bmatrix} -1 & & & \\ 1 & -1 & & \\ & 1 & \ddots & \\ & & \ddots & -1 \\ & & & 1 \end{bmatrix} \tag{1}$$

and the temporal difference signal as:

$$\mathbf{X}\mathbf{D}_h = [\mathbf{x}_2 - \mathbf{x}_1,\ \mathbf{x}_3 - \mathbf{x}_2,\ \ldots,\ \mathbf{x}_M - \mathbf{x}_{M-1}]$$

Now we define the sampling matrix for the whole time-varying graph signal as:

$$\mathbf{J}(i,\ j) = \begin{cases} 1 \ \textbf{if} \ i \in S_t, \\ 0 \ \textbf{if} \ i \notin S_t \end{cases} \tag{2}$$

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As per Giraldo *et al.* (2020), we define Sobolev norm for time-varying graph signals as follows:

$$\|\mathbf{X}\|_{\beta,\epsilon} = \sum_{i=1}^{M} \mathbf{x}_i^{\mathbf{T}} (\mathbf{L} + \epsilon \mathbf{I})^{\beta} \mathbf{x_i} = \mathbf{tr}(\mathbf{X}^{\mathbf{T}} (\mathbf{L} + \epsilon \mathbf{I})^{\beta} \mathbf{X}), \tag{3}$$

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$\mathbf{L}$ being a symmetric matrix.

Finally, we formulate the Sobolev reconstruction problem for time-varying graph signals as:

$$\min_{\overline{\mathbf{X}}} \frac{1}{2} \|\mathbf{J} \circ \overline{\mathbf{X}} - \mathbf{Y}\|_F^2 + \frac{\gamma}{2} \mathbf{tr}((\overline{\mathbf{X}} \mathbf{D}_h)^{\mathbf{T}} (\mathbf{L} + \epsilon \mathbf{I})^{\beta} \overline{\mathbf{X}} \mathbf{D}_h). \tag{4}$$

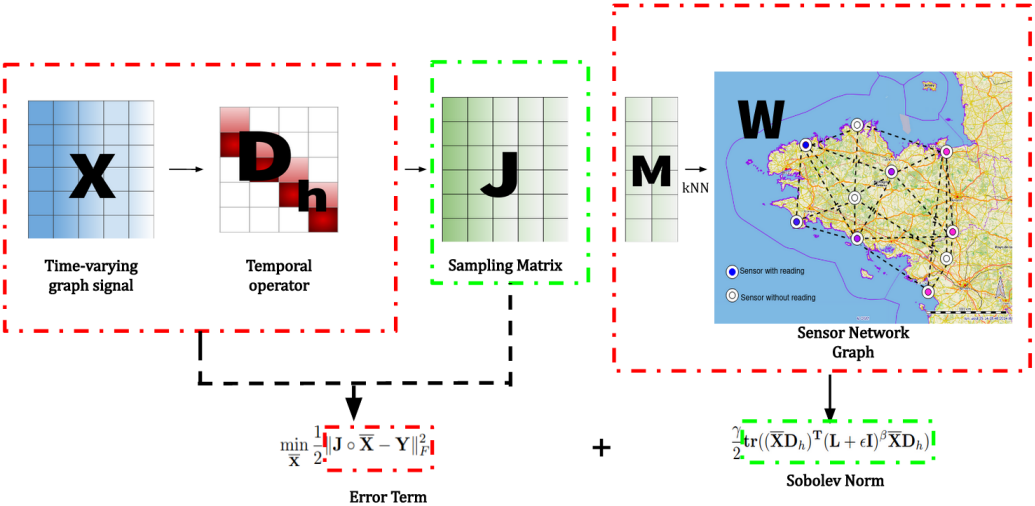

Figure: Schematic diagram.

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Experimental Framework

We evaluate the performance of the proposed method on the following datasets:

- **Moléne Dataset**: An open-access dataset of hourly weather observations in Brittany, France [Girault (2015)].
- **Intel Lab Dataset**: Data collected from 54 sensors deployed in the Intel Berkeley Research Laboratory [Madden (2004)].

We compare the method against SoTA recovery approaches (**kNN** [Kiani et al. (2017)], **EM** [Zhang et al. (2012)], **LRMC** [López-Valcarce et al. (2019)] and **PMF** [Fekade et al (2017)]).

We **randomly remove** some data points for each dataset and then **reconstruct the missing values** using the proposed framework.

Thereafter we compare the missing values with their corresponding original values from the dataset using RMSE and MAE.

	Sampling Density	kNN [2], [3]		EM [4]		LRMC [5], [6]		PMF [7]		Proposed	
		RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE
Molene Dataset	0.1	8.05	6.75	7.34	5.68	3.47	2.35	4.75	2.56	**2.40**	**1.65**
	0.3	5.45	4.68	5.20	3.93	2.82	1.35	3.42	1.64	**1.60**	**1.02**
	0.5	4.20	3.04	4.14	2.95	1.96	1.02	2.45	1.09	**1.20**	**0.74**
	0.7	3.40	2.13	3.19	1.82	1.02	0.81	1.22	0.95	**0.97**	**0.65**
Intel Dataset	0.1	9.70	5.34	7.78	5.63	4.68	2.94	5.34	3.98	**3.04**	**1.50**
	0.3	6.30	3.52	6.45	4.96	3.40	2.15	3.95	2.37	**2.01**	**0.97**
	0.5	4.60	2.98	5.17	3.81	2.06	1.21	2.75	1.65	**1.57**	**0.70**
	0.7	3.75	1.78	4.63	2.57	1.45	0.75	1.97	0.99	**1.30**	**0.55**

Figure: Quantitative comparison of SoTA methods vs proposed method. Best results are in **bold**

Sampling density means the **amount of valid/ available information**, e.g. sampling density of 0.1 means that we have 10% of the data available and we need to reconstruct 90% of the data.

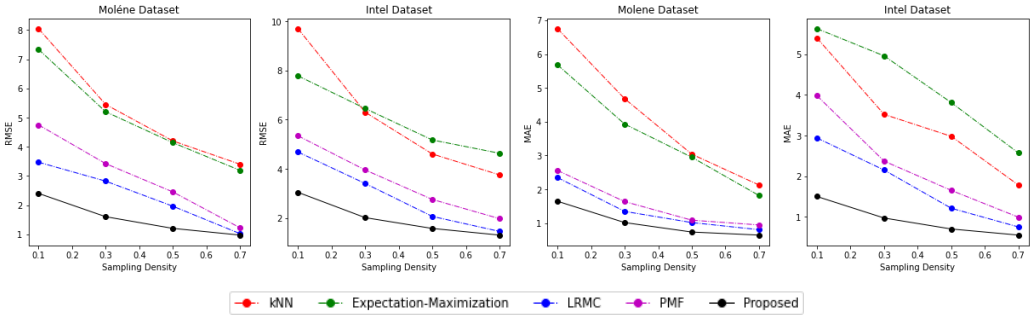

Figure: Performance of various SoTA algorithms in recovering missing sensor data.

This work applies the concept of time-varying graph signals reconstruction to recover missing data in sensor networks.

It leverages the concept of a recently introduced method based on minimization of the Sobolev norm of time-varying graph signals.

This method surpasses multiple previous approaches in the reconstruction accuracy, especially during the unavailability of a big chunk of data.

Future work may include:

- Extending the method to non-smooth situations, e.g. when nearby localities have drastically different temperatures or changes in temperature are abrupt with time.
- To attain this goal, using graph-learning based approaches can be a way forward. We leave this as a future work.

Thank you

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Let us know if you have any questions. Also feel free to contact at anindyam.jan@gmail.com if you are interested in our work and want to explore more!